BRACU Mongol-Toru

Rover Autonomy Recruitment Project

Task 1 : Route Planning

Given Information:

Latitude = 38.418012°

Longitude = −110.777203°

Waypoint 1 (WP1):

Latitude = 38.418603°

Longitude = −110.775784°

Waypoint 2 (WP2):

Latitude = 38.420167°

Longitude = −110.776319°

Waypoint 3 (WP3):

Latitude = 38.420289°

Longitude = −110.776246°

Initial Rover Heading = 0° (North)

Distance Calculation Using Haversine Formula

Start → WP1 distance

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

$$d = 2R \sin^{-1}(\sqrt{a})$$

Where,

$R = 6,371,000$ meters (Earth's radius)

ϕ = Latitude in radius

λ = Longitude in radius

d = Distance in meters.

$\therefore \Delta\phi = \phi_2 - \phi_1 = 0.000591°$

$\therefore \Delta\lambda = \lambda_2 - \lambda_1 = 0.001419°$

Given,

$\phi_1 = 38.418012°$

$\phi_2 = 38.418603°$

$\lambda_1 = -110.777203°$

$\lambda_2 = -110.775784°$

degrees to radians:

$$\Delta\phi = 0.000591° \times \frac{\pi}{180}$$

$$\Delta\lambda = 0.001419° \times \frac{\pi}{180}$$

$\therefore d = 140.01$ meters

WP1 → WP2 distance

$\therefore \Delta\phi = \phi_2 - \phi_1 = 0.001564°$

$\therefore \Delta\lambda = \lambda_2 - \lambda_1 = -0.000535°$

Given,

$\phi_1 = 38.418603°$

$\phi_2 = 38.420167°$

$\lambda_1 = -110.775784°$

$\lambda_2 = -110.776319°$

degrees to radians:

$$\Delta\phi = 0.001564° \times \frac{\pi}{180}$$

$$\Delta\lambda = -0.000535 \times \frac{\pi}{180}$$

$d = 180.05$ meters

WP2 → WP3

Given,

$\phi_1 = 38.420167°$

$\phi_2 = 38.420289°$

$\lambda_1 = -110.776319°$

$\lambda_2 = -110.776246°$

$\Delta\phi = \phi_2 - \phi_1 = 0.000122°$

$\Delta\lambda = \lambda_2 - \lambda_1 = 0.000073°$

degrees to Radians:

$$\Delta\phi = 0.000122 \times \frac{\pi}{180}$$

$$\Delta\lambda = 0.000073 \times \frac{\pi}{180}$$

$\therefore d = 14.98$ meters.

Bearing and Turning Instructions

(1) Bearing Calculation:

Bearing is measured clockwise from North (0°).

The initial heading of the rover is North (0°)

Using the initial heading and the calculated bearings:

- Start → WP1 = 62.01°
- WP1 → WP2 = 345.00°
- WP2 → WP3 = 25.12°

(2) Relative Turning Angles:

Leg1: Start → WP1

Initial heading = 0°

New bearing = 62.01°

Turn = 62.01° RIGHT

Leg2: WP1 → WP2

Previous heading = 62.01°

New bearing = 345.00°

Turn = 62.01° − 345.00° = −282.99°

Taking the shortest turning direction:

Turn = 77.01° LEFT

Leg 3: WP2 → WP3

Previous heading = 345.00°

New bearing = 25.12°

Turn = 25.12° − 345.00° = −319.88°

Taking the shorter turning direction:

Turn = 40.12° Right

(3) Final Rover Command Sequence:

Step	Rover Command
1	Turn RIGHT 62.01°
2	Drive 140.01m
3	Turn LEFT 77.01°
4	Drive 180.05m
5	Turn RIGHT 40.12°
6	Drive 14.98m

GPS Accuracy Analysis (±3 Metres)

(1) GPS Accuracy

The GPS receiver has an accuracy of approximately ±3 metres. This means the actual rover position may differ from the reported GPS position by up to 3 metres.

(2) Effect on Navigation

① The rover may not reach the exact waypoint coordinates.

② Small position errors can affect distance and bearing calculations.

- Errors may become more noticeable when the rover moves between nearby waypoints.
- The rover should use a suitable arrival tolerance instead of expecting an exact coordinate match.

5) Conclusion:

GPS accuracy is important for rover navigation. Since the GPS error can be ±3 metres, the rover should consider a waypoint reached when it is sufficiently close to the target, rather than requiring an exact match.